

CRR Binomial Option Pricing -- AMD \$350 Strike

Forward Pass and Backward Induction Explained

Market Data: AMD collected 2026-05-04 | Expiry 2026-05-22 | American-style put, no dividends

S =	\$341.35	(AMD spot price at collection)
K =	\$350.00	(strike price)
r =	5.3%	(3-month T-bill risk-free rate)
IV =	74.73%	(put implied volatility, solved via CRR solver)
T =	18 calendar days	(0.0493 years)
N =	4 steps	(visual tree) 100 steps (pipeline pricing)

PART 1 -- Forward Pass: Building the Stock Price Lattice

In the forward pass the model builds a recombining binomial lattice of possible AMD stock prices from today (Step 0) out to expiry (Step 4 in the visual, Step 100 in the pipeline). No option values are computed here -- only the underlying stock prices at every node.

CRR Up/Down Multipliers

The size of each price move is derived from the implied volatility:

```
u = exp(IV x sqrt(dt)) = exp(0.7473 x sqrt(0.01233)) = 1.0865
d = 1 / u = 0.9204
dt = T / N = 0.0493 / 4 = 0.01233 years (4.5 days per step)
```

$u = 1.0865$ means AMD moves UP by 8.65% each step. $d = 0.9204$ means AMD moves DOWN by 7.96% each step. Because $d = 1/u$ the tree recombines: an up move followed by a down move lands on the exact same node as a down move followed by an up move. This is why column 2 has 3 nodes instead of 4.

Node Price Formula

Every node (i, j) is computed directly -- no iteration needed:

$$S(i, j) = S \times u^{(i-j)} \times d^j$$

i = time step (column 0 to 4) | j = number of down moves to reach this node | (i-j) = number of up moves

Risk-Neutral Probability

The probability p of an up move is set so that the expected return equals the risk-free rate -- this is the risk-neutral measure:

$$p = (\exp(r \times dt) - d) / (u - d) = 0.4832$$

$$q = 1 - p = 0.5168$$

$p = 0.4832$ and $q = 0.5168$ are NOT real-world probabilities of AMD going up or down. They are mathematical weights used purely to compute fair option values under no-arbitrage pricing. They embed the risk-free rate, not any directional view on AMD.

Terminal Stock Prices at Step 4 (N=4 visual)

Reading the five leaf nodes at expiry (Step 4, 18.0 days):

(4, 0)	4 ups, 0 downs	\$475.7	Far above strike -- deep OTM for put
(4, 1)	3 ups, 1 down	\$403.0	Above strike -- OTM for put

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(4,2)	2 ups,	2 downs	\$341.4	Near spot -- slightly ITM for put
(4,3)	1 up,	3 downs	\$289.2	Below strike -- ITM for put
(4,4)	0 ups,	4 downs	\$244.9	Deep below strike -- deep ITM for put

The spread of these prices -- from \$245.1 to \$475.6 -- is driven entirely by implied volatility (74.73%). Higher IV means larger u and d, which widens the price fan. The vol premium of ~8-9% over RV-30d (65.49%) is what makes these options relatively expensive compared to AMD's recent realized moves.

PART 2 -- Backward Induction: Computing Option Values

Once the full stock price lattice is built, the model works right to left -- from the known terminal payoffs back to the present -- to find the fair value at every node. This is what makes the CRR model an American pricer: at each node it asks whether early exercise is better than waiting.

Step 1 -- Seed Terminal Payoffs (rightmost column)

At expiry the option value is simply the intrinsic payoff. No discounting needed:

$$V(N, j) = \max(K - S(N, j), 0) = \max(350 - S, 0)$$

These seed values feed the backward recursion. Green nodes are ITM ($V > 0$), gray are OTM ($V = 0$).

(4, 0)	S=\$475.7	V=\$0.00	OTM -- gray
(4, 1)	S=\$403.0	V=\$0.00	OTM -- gray
(4, 2)	S=\$341.4	V=\$8.65	ITM -- green
(4, 3)	S=\$289.2	V=\$60.85	ITM -- green
(4, 4)	S=\$244.9	V=\$105.06	ITM -- green

Step 2 -- Recursion Formula at Each Interior Node

For every node (i, j), working from column N-1 back to column 0:

$$\begin{aligned} \text{hold} &= \text{disc} \times (p \times V_{\text{up}} + (1-p) \times V_{\text{down}}) \\ V &= \max(\text{intrinsic}, \text{hold}) \end{aligned}$$

V_{up}	option value at child node (i+1, j) -- already computed
V_{down}	option value at child node (i+1, j+1) -- already computed
p	0.4832 -- risk-neutral probability of up move
disc	0.999347 -- one-step discount factor $\exp(-r \times dt)$
hold	continuation value: what the option is worth if you wait one more step
intrinsic	immediate exercise value: $\max(350 - S, 0)$ at this node

The $\max(\text{intrinsic}, \text{hold})$ is the American exercise decision. If $\text{intrinsic} > \text{hold}$, the model marks the node amber (early exercise optimal) and sets $V = \text{intrinsic}$. If $\text{hold} \geq \text{intrinsic}$, the node is teal (continue holding) and $V = \text{hold}$. European options skip this comparison and always use hold.

Step 3 -- Node Color Key and Decision Rule

Amber node	$\text{intrinsic} > \text{hold}$ -- exercise now is optimal; $V = \text{intrinsic}$
Teal node	$\text{hold} \geq \text{intrinsic}$ -- wait is optimal; $V = \text{continuation value}$
Green node	terminal ITM -- $V = \max(K - S, 0) > 0$ (expiry payoff)
Gray node	terminal OTM -- $V = 0$ (option expires worthless)

Step 4 -- Worked Example: Step 3 Nodes

Reading the four nodes at Step 3 (13.5 days in) from the animation:

(3, 0)	S=\$437.8	intr=\$0.00	hold=\$0.00	teal	OTM -- no intrinsic value
(3, 1)	S=\$370.9	intr=\$0.00	hold=\$4.47	teal	OTM but time value remains
(3, 2)	S=\$314.1	intr=\$35.86	hold=\$35.58	amber	Intrinsic beats hold -- exercise now
(3, 3)	S=\$266.3	intr=\$83.67	hold=\$83.57	amber	Intrinsic beats hold -- exercise now

At node (3,2): intrinsic \$35.86 vs hold \$35.58 -- early exercise wins by \$0.28. At node (3,3): intrinsic \$83.67 vs hold \$83.57 -- early exercise wins by \$0.10. Both are cases where the put is deep enough in the money that the time value of

waiting is worth less than collecting the intrinsic payoff immediately.

Step 5 -- Root Node: The Final Option Price

After all columns are processed, the root node (0,0) holds the model price:

```
V(0,0) = $27.0739 (N=4 visual tree)
V(0,0) = $27.0014 (N=100 pipeline tree)
V_market= $27.0000 (Fidelity limit price)
Error = +0.005% (model vs market)
```

With only N=4 steps the tree approximates \$27.08. With N=100 steps the binomial tree converges to \$27.0014 -- matching the Fidelity market price within 0.005%. The small residual is floating-point rounding from IV calibration, not a model deficiency. This accuracy confirms the CRR model is correctly implemented and the implied volatility was solved precisely.

Why the Model Price Matches Market So Exactly

The put IV (74.73%) was solved by running `scipy.optimize.brentq` against the CRR pricer: it searched for the sigma value that makes $V_{\text{model}} = V_{\text{market}} = \27.00 . Because the IV was calibrated to the market price, the model reproduces that price by construction -- the near-zero edge is expected, not a coincidence. To find genuine mispricing edge, an independent IV forecast (e.g., forecasted realized volatility) must be used instead of market-implied IV.

SUMMARY: Forward Pass vs Backward Induction

Property	Forward Pass	Backward Induction
Direction	Left to right (Step 0 to N)	Right to left (Step N to 0)
Computes	Stock price S at each node	Option value V at each node
Formula	$S(i,j) = S \times u^{(i-j)} \times d^j$	$V = \max(\text{intrinsic}, \text{disc} \times (p \cdot V_u + q \cdot V_d))$
Node color	All blue (stock price nodes)	Amber/teal/green/gray (see legend)
Exercise?	N/A -- no option decision yet	Yes -- American early exercise at each node
Seeds	$S_0 = \$341.35$ at root	Terminal payoffs $\max(K-S, 0)$ at leaves
Key output	Terminal price fan	$V(0,0) = \text{model option price}$
N=4 result	5 terminal prices \$245-\$476	$V(0,0) = \$27.07$
N=100 result	101 terminal prices	$V(0,0) = \$27.0014$ (error 0.005%)

Animation: `crr_binomial_pricing_amd_html.py` | Output: `crr_pipeline_animation.html` | CS495 Deep Scholar, Bellevue College Spring 2026

PART 3 -- V_market and V_model: Meaning and Interpretation

Every ticket in the pipeline produces two prices: V_market and V_model. Understanding the difference between them -- and what their gap means -- is the core of the mispricing edge framework.

V_market -- The Real-World Transaction Price

V_market is the price you actually pay (or receive) in the real world. It is the limit price on the Fidelity order ticket -- the amount of cash that changes hands per share when you buy or sell the option. For the AMD \$350 tickets:

```
Call V_market =    $18.95  -- cost $1,895 per contract (100 shares x $18.95)
Put  V_market =    $27.00  -- cost $2,700 per contract (100 shares x $27.00)
```

V_market reflects everything the market collectively believes: supply and demand, fear and greed, upcoming events, and the market's implied volatility. It is the ground truth of what the option costs right now. No model is required to observe it -- it is simply read from the order book.

V_model -- The CRR Theoretical Fair Value

V_model is what the CRR binomial tree computes the option should be worth, given the inputs you feed it (S, K, r, IV, T, N). For the AMD \$350 tickets:

```
Call V_model =    $18.9486  (pipeline N=100, Call IV=73.92%)
Put  V_model =    $27.0014  (pipeline N=100, Put  IV=74.73%)
```

V_model is a theoretical fair value derived purely from mathematics -- backward induction through the binomial lattice, discounting expected payoffs under the risk-neutral measure. It is the model's answer to the question: given these inputs, what is this option worth?

The Edge -- Interpreting the Gap

The mispricing edge measures how far V_model deviates from V_market:

```
edge = (V_model - V_market) / V_market
```

Condition	Edge Sign	Interpretation
$V_model > V_market$	Positive edge	Model says option is UNDERPRICED by market -- potential buy signal
$V_model < V_market$	Negative edge	Model says option is OVERPRICED by market -- potential sell signal
$V_model = V_market$	Zero edge	No detectable mispricing -- no edge, no trade recommended

For the AMD \$350 tickets the edge is -0.007% (call) and +0.005% (put) -- effectively zero. Both fall well below the 2% minimum threshold configured in the pipeline, so the Kelly Criterion outputs zero position size and recommends NO TRADE for all four tickets.

Why the Edge Is Near Zero for These Tickets

The near-zero edge is not a failure of the model -- it is a consequence of how IV was obtained. The implied volatility (73.92% call, 74.73% put) was solved by running `scipy.optimize.brentq`: searching for the sigma value that makes $V_model = V_market$ exactly. When you calibrate the model to the market price, the model reproduces that price by construction. There is no independent prediction being made; the model is confirming the price it was given.

```
IV solved from market      => V_model = V_market      => edge ~ 0%
IV forecast independently => V_model may diverge    => edge != 0%
```

To find genuine mispricing edge, you must supply an independent IV estimate -- for example, a forecasted realized

volatility based on historical data, a statistical model, or a macro view. If your forecast IV differs from the market-implied IV, the CRR model will produce a V_{model} that diverges from V_{market} , and that divergence becomes the edge the Kelly Criterion sizes around.

Practical Interpretation Example

Suppose you forecast AMD's realized volatility over the next 18 days will be 65% (close to the RV-30d of 65.49%) rather than the market-implied 74.73%. Running the CRR pricer with IV=65% instead of 74.73% would produce a lower V_{model} for the put:

```
V_model (IV=74.73%) = $27.00 -- matches market, edge = 0%
V_model (IV=65.00%) = ~$21.50 -- below market, edge ~ -20%
Interpretation: put is OVERPRICED by ~20% relative to your vol forecast
Kelly signal: SELL the put (collect the vol premium)
```

The market is charging \$27.00 for a put you believe is worth only ~\$21.50 based on your volatility forecast. That \$5.50 gap -- the vol premium -- is the edge you would be collecting if you sold the put. Whether to act on it depends on confidence in your forecast and the Kelly fraction sizing.

QUICK REFERENCE: V_{model} vs V_{market}

Term	Formula / Source	What It Means
V_{market}	Real-world price	Cash paid/received at Fidelity order fill
V_{model}	Model fair value	CRR backward induction result given inputs
Edge	$(V_m - V_k) / V_k$	Fractional mispricing; drives Kelly sizing
Edge = 0	IV from market	Model calibrated to market; no prediction made
Edge != 0	IV independent	Genuine divergence; Kelly can size a position
Min edge	2% threshold	Pipeline filters noise; below 2% = NO TRADE

Animation: [crr_binomial_pricing_amd_html.py](#) | Output: [crr_pipeline_animation.html](#) | CS495 Deep Scholar, Bellevue College Spring 2026